Dustin Biser

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Interests and Focus

Computer Graphics, Collision Detection, Rigid Body Simulation, and Numerical Methods.

Education

University of Maryland, College Park
Bachelor of Science in Physics, 2012
Bachelor of Science in Mathematics, 2012
Minor in Computer Science

Recent Projects

3D Graphics Engine Link
July 2013 – Present
Engine for real-time 3D graphics. Developed using C++, OpenGL, and GLSL.

Physics Engine Link
December 2011 – May 2012
My senior project in college, which involved creating a 2D physics engine for convex rigid bodies.

Employment

Software Engineer II
May 2013 – September 2013
Appian (Reston, VA)
Market leading company focused on building cutting edge Business Process Management (BPM) applications and tools.

- Java development in fast pace agile environment aimed at creating BPM tools for businesses in order to boost their overall efficiency.
- Designed, implemented, and tested new features for cutting edge web and mobile BPM applications.

Computer Vision Software Engineer
June 2012 – March 2013
SAIC (Arlington, VA)
Government contractor focused on a mix of research and building sophisticated software applications for the military and the Department of Defense.

- C++, OpenGL, Qt development on massively simulated 3D graphical application used for military planning and training.
- Java, OpenGL development on multithreaded application used for stabilizing and stitching together video frames from unmanned aerial vehicles in real time.

Web Programmer
May 2010 – Aug 2011
Optoro (Lanham, MD)
Company focused on asset recovery and product management. They create cloud based tools to allow companies to easily track, manage, and sell merchandise online.

- Ruby on Rails development on cloud based product management system used for receiving, selling, and shipping online merchandise.
- Lots of work involving Ruby scripting, Databases, Git, Agile Development, and Test Driven Development.

Research Experience

Undergraduate Research Intern
May 2009 – August 2009
National Institute of Standards and Technology (Gaithersburg, MD)
Developed an interactive 3D graphical solution for interpolating and displaying the amounts of radiation delivered throughout containers that had been sent through industrial irradiation facilities.

Undergraduate Research Intern
May 2008 – August 2008
National Institute of Standards and Technology (Gaithersburg, MD)
Created and validated a software controller for programming electro-optical devises used in high-fidelity remote sensing instrument calibrations.

Main Skills

- Strong 3D graphics theory knowledge
- Strong Math and Physics knowledge
- Excellent written and spoken communication skills
- Experience with multithreaded programming and architectures
- Modeling: 3D Mesh Creation and UV Mapping
- Fluent in C/C++, GLSL, and Java.
- Libraries: OpenGL, Boost, STL, GLM, Qt, GoogleTest
- Graphics Tools: Blender, Gimp, Photoshop
- Dev. Tools: Visual Studio, Eclipse, git, subversion, premake4, cmake

Open-Source Software Contributions

OpenGL Mathematics (GLM) Link
October 2013 – Present
So far I have contributed bug fixes and optimizations for various quaternion and matrix routines. I plan to continue contributing to this project in the future as it is very useful and has gained much popularity within the graphics community.